

Gonçalo Magno

Generalist software engineer · agentic engineering

Brazil and Portugal · remote (GMT-3 / GMT)

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SUMMARY

Generalist software engineer with over a decade of experience designing, building and running cloud-native systems, mostly in Python on AWS. Particularly strong in observability (logging, metrics and tracing with Datadog, Dynatrace and OpenTelemetry), public APIs and MongoDB, and in agentic engineering: bringing AI into products and into the way teams build software. Comfortable owning services end to end, from design to production. Has lived and worked on three continents (UK, Portugal, Angola, Brazil), often in multicultural teams. European citizen, fluent in English and Portuguese.

EXPERIENCE

Senior Software Engineer, VASS

2024–present

Remote · client TELUS Health

User Management and Onboarding team, which owns how client organisations and their members are onboarded onto a wellbeing platform.

- Designed and built a public Organisations and Users API from inception to general availability (async FastAPI and Pydantic on ECS Fargate and Lambda, event-sourced over MongoDB and DynamoDB, Terraform); it is the platform's programmatic onboarding channel for enterprise clients across four regions.
- Delivered standards-first SCIM 2.0 provisioning (RFC 7643/7644) with PATCH, filtering, bulk operations and cross-region identity uniqueness; implemented and tested separate integrations with Microsoft Entra ID and Okta, and passed Microsoft's SCIM validator.
- Conceived and built, end to end, an AI assistant for customer-support agents, now in production: it streams answers from a Claude-based copilot with retrieval and MCP tools through a hardened FastAPI streaming gateway and an accessible React widget, with SSO, fail-closed access control, per-user rate limits, a spend kill switch and test-enforced PII/PHI-free logging.
- Hands-on in the Datadog-to-Dynatrace migration: OpenTelemetry over OTLP for containerised (ECS) workloads, W3C trace context propagated across Step Functions and the OneAgent layer for Lambda, with a dual run for a zero-gap cutover and the tagging standard encoded once and reused.
- Identified the cause of nearly half of the production CloudWatch bill (a monitoring integration polling metrics in every AWS region instead of the few in use) and got it fixed the next day.
- Raised the engineering baseline across the team's Python services: modern tooling (uv), pre-commit with Ruff and strict type checking, code and dependency scanning, keyless OIDC deployments and automated semantic-versioned releases on GitHub Actions and CircleCI.
- Built the team's shared AI tooling (an AI pull-request reviewer packaged as a reusable action, plus per-repository agent instructions and guardrails), making agentic development part of the everyday workflow.
- Helped design the next-generation onboarding pipeline and built two of its core components: strict Pydantic-based validation that rejects malformed client files before they reach the platform, and a diff engine that works out exactly which members to add, update or remove.
- Recognised on several occasions by leadership and peers; a regular presenter at the fortnightly engineering-wide demo, and an interviewer in technical hiring, helping bring several engineers onto the team.

Senior Software Engineer, WebOps Tech

2018–2024

Remote

Own consultancy (web-ops.tech), delivering backend engineering for scale-ups in fintech and web3, e-commerce and mobility.

- **Zharta** (on-chain lending, web3): built serverless services on AWS Lambda, DynamoDB and Terraform; introduced CloudWatch dashboards and alarms as code routed to Slack; removed deployment downtime by rolling out Lambda aliases; introduced dependency injection to make the services testable.
- **Kencko** (e-commerce): worked on Python services (FastAPI and Flask, SQLAlchemy) backed by PostgreSQL and SQLite, on Kubernetes; extracted logic from the monolith into a new FastAPI service; owned integrations with the Shopify GraphQL API, shipping carriers, Contentful and SendGrid.
- **Rydes, now NAVIT** (mobility): built payouts and a custom invoicing mechanism in NestJS and PostgreSQL on GKE; integrated digital banking and card payments, and CO₂ offsetting.

Software Developer, Unity Technologies

2017–2018

United Kingdom

- Designed and built a device booking and provisioning system spanning many gaming platforms, operating systems and architectures, giving developers cloud-like access to physical test devices.

Software Engineer, Imagination Technologies

2012–2016

United Kingdom

- Led a small team building an in-house provisioning automation framework in Python (Django).

EDUCATION

MSc Electronics and Telecommunications Engineering, University of Aveiro

2002–2008

Integrated master's degree, Portugal (BSc 2006, MSc 2008).

SKILLS

Programming languages: Python, SQL, TypeScript/JavaScript, Bash, C

Cloud & infra: AWS (Lambda, Step Functions, ECS Fargate, SQS, EventBridge, DynamoDB, Cognito), GCP, Terraform, AWS CDK, Docker, Kubernetes

Data: PostgreSQL, SQLite, MongoDB, DynamoDB

Observability: OpenTelemetry, Dynatrace, Datadog, CloudWatch; logging, metrics and tracing

AI / agentic engineering: LLM-backed product features (Claude, RAG, MCP), AI code review, agentic development workflows

Frameworks and standards: FastAPI, Pydantic, SQLAlchemy, SCIM 2.0, OAuth 2.0

CI/CD and tooling: GitHub Actions, CircleCI, pre-commit, Ruff, mypy, pytest, uv

Practices: CI/CD and release automation, testing, code review, technical writing

SELECTED OPEN SOURCE

- **spyci**: Python parser for SPICE raw simulation files.
- **tmon**: command-line CPU temperature monitor.
- **igl**: OpenGL bitmap image viewer in C.
- **vds**: semantic search over text documents.

LANGUAGES

Portuguese: Native

English: Fluent